



ENVIRONMENT

SOCIAL ROBOT CO-DESIGN CANVASES

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

ED waiting/reception area, before nurse assessment; connects securely to the hospital network and EHR.

Where

What place?
Does it change?

Emergency Department waiting or reception area, positioned before nurse assessment. Placed to maintain patient privacy while remaining visible and accessible.

User(s)

Who is using the robot?

Emergency Department patients arriving for self-check-in and preliminary triage data collection.

When

What time of day?
Does it change?

Continuous use during all opening hours, day and night; busiest during peak ED demand. Built with durable, easy-to-clean materials for continuous daily use and infection control.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

ED reception and triage staff, who receive escalations; clinical staff who respond to abnormal readings or technical faults.

Data collection

Does the robot collect data from its environment?
How is it stored?

Collects demographic information, presenting complaint, and vital signs from integrated medical devices; securely transmitted to the Mercer EHR.



TRADE-OFF:

More data collection requires more attention to data security.

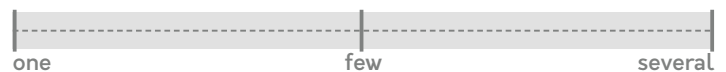
Simultaneous users

How many users should be able to use the robot simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



One patient per kiosk at a time; guides a single patient through self-check-in and preliminary assessment before nurse triage.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

Sensors: blood pressure cuff, pulse oximeter, thermometer, barcode/ID scanner, microphone, camera (where approved). Outputs: touchscreen display, speaker, visual/audio status indicators.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Connects securely to the hospital network, Mercer EHR and nurse dashboard; sends AI-generated preliminary ESI recommendations directly to triage staff.



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THRI paper:

